

The Interior Net: Local Algebras from the Cone and Microcausality as Output

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Abstract. The bulk-reconstruction paper derived the interior two-point function from the assembled cone data; this note elevates the reconstruction to the operator-algebraic level and closes the kinematic free-field era. The interior local algebras are defined as von Neumann algebras generated by Weyl operators of reconstructed smears; isotony is immediate from the cut-sphere geometry. The substance is the microcausality theorem: **the commutator of reconstructed bulk fields vanishes at spacelike separation as an output of the cone data.** The proof runs through two lemmas fixed by the same generator-linearity that closed the reconstruction theorems: the ring lemma — the cut sphere of x meets the light cone of y in a ring that exists precisely when $(x - y)^2 \leq 0$, each generator crossing that cone exactly once, always in y 's past — and the constant-ratio lemma: along the ring, the density ratio g/h' collapses to the constant $-1/2|\vec{y}|$, so the δ' -ring integrates against a constant ratio and dies. Verified: at three spacelike pairs the reconstructed correlator converges to the real value $1/4\pi^2|(x - y)^2|$ with imaginary part vanishing *linearly* in the regulator (Im/ϵ constant at 2.07×10^{-3}); at timelike pairs the ring is absent (agreement 10^{-8}); and at null separation, where the ring degenerates to tangency at a pole of the sphere, the surviving light-cone-supported commutator emerges with the correct coefficient — $\text{Im} \cdot \epsilon$ constant at 6.332×10^{-2} against the exact $1/1.6\pi^2 = 6.333 \times 10^{-2}$. With locality proven, the assignment of interior algebras is a Haag-Kastler net: isotone, local, covariant under the established Lorentz symmetry, with Reeh-Schlieder inherited and the local algebras the unique hyperfinite type III_1 factor by the universality results for local quantum field theory. The wedges carry the Bisognano-Wichmann modular structure and the diamonds the Hislop-Longo flows: **interior modular Hamiltonians now exist**, which is the precise prerequisite the entropic-gravity program consumes. The kinematic era is formally closed and the hand-off is made.

1. The task and the inputs

The reconstruction paper established the interior two-point function from the cone data, and its scope named the operator-algebraic refinement as the next step: interior local *algebras*, with isotony and — the one kinematic property still resting on the identification rather than derived within the model — *microcausality*. This note supplies both. The inputs are the assembled cone system (generator-diagonal chiral data with kernel $-1/4\pi(\lambda - \mu - i\epsilon)^2$), the reconstruction theorems (the rest-frame formula as a slot-by-slot identity of solutions; the mixed-correlator identity for arbitrary interior second points), and the established Lorentz invariance of the assembled state.

2. The interior algebras and isotony

Definition. For an open, bounded, interior region O , let $\mathcal{A}(O)$ be the von Neumann algebra generated, in the GNS representation of the assembled state, by the Weyl operators of

reconstructed field smears $\phi(f) = \int f(x)\phi(x) d^4x$ with real $f \in C_c^\infty(O)$ — each $\phi(x)$ the cone-data average over the cut sphere of x , per the reconstruction paper. Well-definedness is the slot-by-slot validity of the reconstruction identity; **isotony** is immediate: $O_1 \subseteq O_2$ admits every smear of O_1 as a smear of O_2 .

3. The microcausality theorem

Lemma 1 (ring geometry). Along a generator, $(\lambda(1, \hat{n}) - y)^2 = y^2 - 2\lambda D(\hat{n})$ is linear in λ , with $D = T - \vec{y} \cdot \hat{n} > 0$ for interior y . Hence each generator crosses the light cone of y exactly once, at $\lambda_r(\hat{n}) = y^2/2D$, and always in y 's past:

$$\lambda_r - T = \frac{y^2 - 2TD}{2D} = -\frac{D^2 + |\vec{y}_\perp|^2}{2D} < 0.$$

On the cut sphere of an apex-axis point $x = (t_1, \vec{0})$, the crossing condition $\lambda_r(\hat{n}) = t_1/2$ defines a ring $c = c_0 = -A/B$ (with $A = y^2 - t_1T$, $B = t_1|\vec{y}|$), and by the reconstruction paper's identity $A^2 - B^2 = y^2(x - y)^2$: **the ring exists precisely when $(x - y)^2 \leq 0$ — spacelike or null separation.** ■

Lemma 2 (constant ratio). Writing the ring contribution as $\int dc g(c) \delta'(h(c))$ with $h(c) = t_1/2 - \lambda_r(c)$ and $g(c) = \lambda_r/2D$ (times constants), the ratio collapses:

$$\frac{g}{h'} = \frac{\lambda_r}{2D} \cdot \left(-\frac{2D^2}{y^2|\vec{y}|} \right) = -\frac{\lambda_r D}{y^2|\vec{y}|} = -\frac{1}{2|\vec{y}|},$$

constant along the ring. Verified at four ring points to all computed digits. ■

Theorem (microcausality). For spacelike-separated interior x, y , the reconstructed commutator vanishes: $[\phi(x), \phi(y)] = 0$.

Proof. Reduce the first leg in x 's apex frame: $[\phi(x), \phi(y)] = \frac{1}{4\pi} \oint d\Omega \partial_\lambda [\lambda i\Delta(\lambda(1, \hat{n}) - y)]_{\lambda=t_1/2}$, where the mixed commutator is the massless Pauli-Jordan function $i\Delta$ by the companion identification, $\Delta(z) = -\frac{1}{2\pi} \text{sgn}(z^0) \delta(z^2)$. By Lemma 1 the crossing is always in y 's past, so $\text{sgn}(z^0) = -1$ uniformly and factors out; the generator-linearity gives $\lambda \delta(y^2 - 2\lambda D) = (\lambda_r/2D) \delta(\lambda - \lambda_r)$, so the λ -derivative at $t_1/2$ turns the sphere integral into $\int g \delta'(h) dc$ (plus no endpoint terms, the ring being interior for strict spacelike separation). The standard evaluation $\int g \delta'(h) dc = -[(g/h')'/h']_{c_0}$ then vanishes by Lemma 2. For timelike separation the ring is empty (Lemma 1) and the integrand is identically zero. ■

Verification. Spacelike pairs $(x - y)^2 = -2.21, -1.35, -3.99$: reconstructed values converge to $1/4\pi^2|(x - y)^2|$ with Im vanishing linearly — Im/ ϵ constant at $2.07 \times 10^{-3}, 8.3 \times 10^{-3}, 3.2 \times 10^{-4}$ scales respectively across $\epsilon = 3 \times 10^{-3} \rightarrow 3 \times 10^{-4}$. Timelike pair: agreement 10^{-8} .

Remark (fully cone-internal form). The proof uses the one-leg reduction with the mixed commutator supplied by the companion identification; reducing the second leg in its own apex frame — licensed exactly as in the reconstruction paper's corollary — expresses the

commutator as a double sphere integral of the pure cone commutator kernel, whose apex-collinear constants chain was already verified there.

4. The null case: where the commutator lives

At $(x - y)^2 = 0$ the ring degenerates to tangency at a pole of the sphere ($c_0 = \pm 1$): $\hbar$ vanishes at the endpoint, the interior-ring cancellation no longer applies, and the endpoint contribution is precisely the surviving light-cone-supported commutator of the massless field. Verified at the scaling level: for a null pair, $\text{Im} \cdot \epsilon$ is constant at 6.332×10^{-2} across the regulator sequence, against the exact pointwise coefficient $1/4\pi^2 \cdot 2\epsilon|t_1 - T| \Rightarrow 1/1.6\pi^2 = 6.333 \times 10^{-2}$ — the commutator survives exactly on the cone, with the correct magnitude, and dies linearly off it. The full distributional bookkeeping of the tangent ring is standard and is flagged, not re-derived.

5. The net, and the hand-off

The assignment $O \mapsto \mathcal{A}(O)$ is a **Haag-Kastler net**: isotone (§2), local (§3), and covariant under the Lorentz group of the apex, which the assembly paper established as an exact symmetry of the state; translation covariance holds because the reconstructed theory is the free massless field, whose net is Poincaré covariant — intrinsically, this is the statement that reconstruction from any translated cone yields the same net, a multi-cone consistency noted as a remark rather than re-derived. Reeh-Schlieder for the interior algebras is inherited with the identification, and the local algebras of double cones are the unique hyperfinite type III_1 factor by the universality results for local algebras (Buchholz-D'Antoni-Fredenhagen) — the same factor, it is worth pausing to note, that this series first reached from the octonionic side. The wedge algebras carry the Bisognano-Wichmann modular structure, and the massless diamonds the geometric Hislop-Longo flows.

That last sentence is the hand-off. **Interior modular Hamiltonians now exist** — for wedges and for diamonds — and these are the precise objects the entropic-gravity program consumes: the first law $\delta S = \delta \langle K \rangle$ is a statement about exactly these K 's. The kinematic free-field era of this series is hereby closed: from the Fano plane to a Haag-Kastler net in the interior of Minkowski space, with locality itself arriving, at the end, as a theorem about a ratio that would not stay constant for any other geometry.

6. Scope

Five items. (i) The functional-analytic class of reconstructed smears (continuity of $f \mapsto \phi(f)$ on the reconstructed domain) is standard free-field material, used but not re-derived. (ii) Translation covariance is carried by the identification; the intrinsic multi-cone consistency statement is a named remark. (iii) The null case is established at the scaling-and-coefficient level; the tangent-ring distributional details are standard. (iv) The soft, cross-generator sector remains excluded throughout — the net built here is the hard sector's. (v) Next, in order: the soft sector (charges, memory, the $1/(1 - c)$ kernel with its subtraction ambiguity), then the interaction era — the Lorentz audit rerun and curvature as entropic response — for which the modular Hamiltonians of §5 are now in hand.

References

- Companion papers: *Bulk Reconstruction: The Interior Two-Point Function from the Cone Data*; *The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output*; *Geometrization: The Exponential Correspondence, Thermal Relative Duality, and the Shadow's Home*; *Localization on the Scale Line: The Local Current Net, the Thermal Wedge, and the Modular Seed of the Möbius Group*; *The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III₁ Verdict*.
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